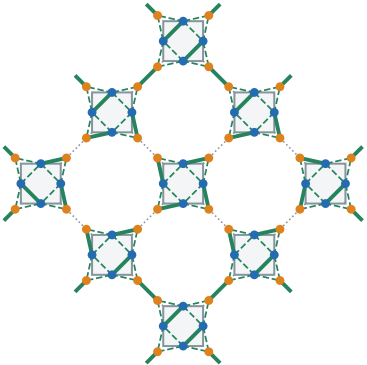
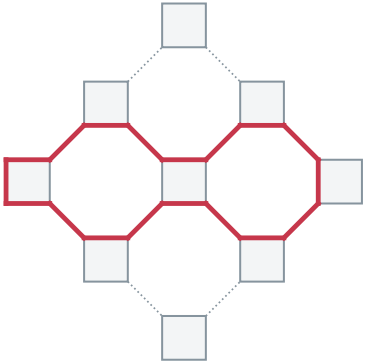
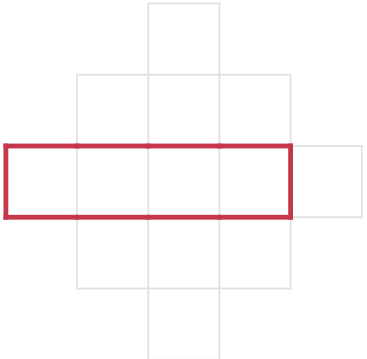
1 Singlet covering



2 Reference-relative loops



3 Contract dummy edges